

ECO204: Project 2, Merck & Company Case

Hanum Sukma Zahrani (1010180340)

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Professor Ajaz

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Introduction

Merck & Co., Inc. must decide whether to license Davanrik, a new drug developed by LAB Pharmaceuticals to treat both depression and obesity. LAB Pharmaceuticals does not have the resources to complete the clinical approval process, manufacture the drug, and market it. The company has approached Merck with an offer to license the compound. Under this agreement, Merck will handle regulatory approval, production, and marketing, while LAB will receive an initial payment, milestone payments, and royalties on future sales.

To determine whether Merck should license Davanrik, I analyze the decision in two parts. In Part A, I build a decision tree to calculate the expected value of licensing, which involves comparing the expected value of completing the FDA approval process and launching the drug versus the expected value of not licensing it. This analysis is based on known probabilities and financial projections. In Part B, I introduce uncertainty into some key parameters and run Monte Carlo simulations to assess how this uncertainty affects the decision. This approach helps visualize real-life conditions where predictions are not always accurate. By incorporating an additional probability factor, I control for cases where predictions may either underestimate or overestimate potential outcomes. I will assume Merck as a risk-neutral decision-maker, thus I will base the optimal decision solely on the expected value.

Executive Summary

This report evaluates Merck's decision on whether to license Davanrik by assessing its financial viability through both deterministic and probabilistic analyses. The licensing decision requires Merck to enter a three-phase clinical approval process, which involves significant costs and uncertainty. To determine whether licensing is an optimal decision, I first construct a decision tree to calculate the expected value of licensing compared to not licensing. The decision tree analysis shows that licensing Davanrik has an expected value of \$13.98 million, making it a favorable investment under the given assumptions.

Since real-world conditions involve uncertainty, I incorporate uncertainty into key parameters such as success probabilities and costs to create a more realistic model. Given this uncertainty, I conduct Monte Carlo simulations with 1,000 to 5,000 iterations to examine how different assumptions affect the decision. The results indicate that after controlling key parameters, the frequency of choosing to license declines, demonstrating that different assumptions lead to different optimal choices. In some cases, licensing remains the best option, while in others, doing nothing becomes the more favorable decision. However, the average expected value varies but remains positive across simulations. For 1000 - 3000 simulations the average expected value consistently falls within the \$15M to \$17M and it becomes even more consistent to only around \$16M for 4,000-5,000 simulations. This confirms that, overall, licensing Davanrik remains the optimal decision, as it continues to generate positive value despite accounting for uncertainty.

Optimal Decision

Merck's decision to license Davanrik hinges on a thorough evaluation of potential revenue, clinical trial success probabilities, and the associated development costs. The development of Davanrik involves three critical phases of clinical trials before the drug can be approved and marketed. At each stage, there is a

risk of failure, and the company must assess whether the expected revenue from launching Davanrik outweighs the costs and risks of continuing the trials.

The final decision also takes into account the competitive landscape and regulatory hurdles that could impact the drug's market success. Since several of Merck's best-selling drugs are nearing patent expiration, securing a new source of revenue is essential for maintaining financial stability. If Davanrik is successful, it could provide a competitive advantage and offset potential revenue losses from expiring patents. However, if the drug fails at any stage, Merck will face significant financial losses with no return on its investment. Given these factors, the decision to license Davanrik depends on whether the potential long-term revenue justifies the high costs and risks associated with its development.

Methodology

Part A: Decision Tree with Backward Induction

To determine the optimal decision, I visually represented all possible outcomes using a decision tree, as shown in Figure 1 below. This tree graph includes both the success and failure probabilities for each outcome. In the decision tree In the analytical process, I use backward induction to calculate the expected value of licensing Davanrik. Backward induction is a systematic method that starts from the final stage of the decision tree and works backward to determine the best decision at each preceding stage. This approach ensures that Merck makes the most rational and financially sound decision by considering future outcomes before committing to earlier investments.

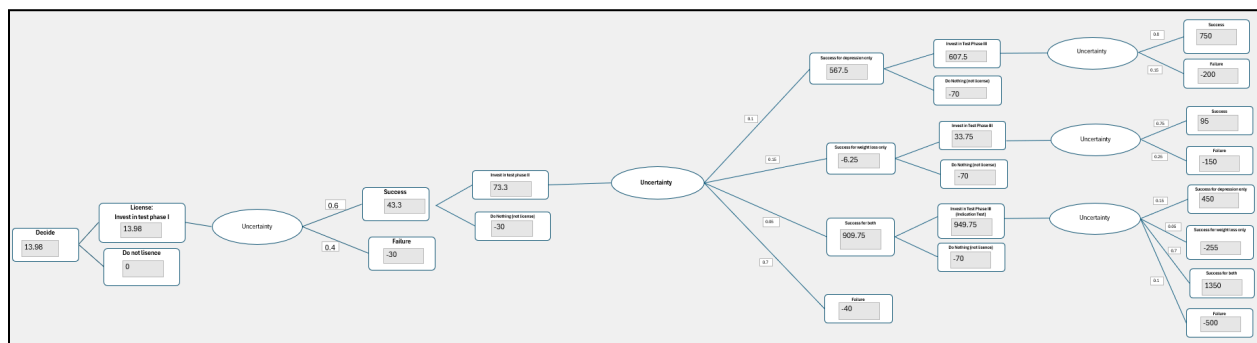


Figure 1

The process begins at Phase III, where the drug undergoes large-scale clinical trials to confirm its safety and effectiveness. If the drug reaches this stage, its probability of success depends on which indication was successful in Phase II. If the drug was effective for depression only, there is an 85% probability of Phase III success. If it was effective for weight loss only, the success probability is 75%. If it showed effectiveness for both conditions, additional specialized trials are required, leading to higher costs and a 70% success probability. In cases where Phase III fails, Merck incurs significant financial losses, as all previous investments in the drug's development would be wasted.

Using backward induction, I calculate the expected value at Phase III by weighing the potential revenue and losses against their respective probabilities. If Phase III succeeds, the drug generates revenue based on its approved indication. If it is approved for depression only, Merck earns \$1.2 billion in revenue,

leading to a net profit of \$950 million. If approved for weight loss only, revenue is \$345 million, resulting in a net profit of \$245 million. If the drug is approved for both conditions, revenue reaches \$2.25 billion, yielding a net profit of \$1.85 billion. These profits are discounted by their respective probabilities to compute the expected value of investing in Phase III.

Moving backward to Phase II, the drug is tested on a larger patient sample to assess its effectiveness. There are four possible outcomes: success for depression only (10%), success for weight loss only (15%), success for both conditions (5%), or failure (70%). If the drug fails, Merck loses its Phase II investment with no return. If it succeeds in any category, the expected value of proceeding to Phase III is derived from the calculations already performed in the previous step.

At Phase I, the drug is tested on a small group of healthy volunteers to evaluate its safety. This stage has a 60% probability of success and a 40% probability of failure. If the drug fails in Phase I, Merck loses the initial \$30 million investment and cannot proceed further. If it succeeds, the expected value of proceeding to Phase II is determined by considering the potential success and failure probabilities from the next stage.

Finally, at the initial decision point, Merck must choose between licensing Davanrik or rejecting the offer. If Merck chooses not to license, they will get \$0, meaning no financial gain or loss. If Merck chooses to license the drug, the expected value of this decision is calculated using all probabilities and financial values derived from the backward induction process. The decision tree analysis shows that, under the assumed probabilities and cost structure, licensing produces a positive expected value of \$13.98 million. This means that, based purely on deterministic expected values, licensing Davanrik is the optimal decision. However, these calculations assume fixed probabilities and costs, which is often not realistic in pharmaceutical development.

Part B: Incorporating Uncertainty Using Monte Carlo Simulations

To account for real-world unpredictability, I introduced Monte Carlo simulations, which model how different scenarios might unfold based on varying trial success rates and costs. Unlike the deterministic model, which assumes fixed probabilities and financial projections, this approach allows these values to change within a reasonable range to simulate different possible outcomes.

When setting the thresholds for randomness, I made several reasonable assumptions based on industry trends and factors that commonly affect pharmaceutical trials. The success probabilities of Phase I and Phase II trials may fluctuate due to differences in patient response, unforeseen side effects, or regulatory scrutiny. Therefore, instead of assuming a strict 60% success rate for Phase I, I allowed it to vary between 40% and 80% to reflect possible trial variations. Similarly, for Phase II, where the drug could be effective for depression, weight loss, or both, I introduced a range of 0% to 10% for both conditions, acknowledging that trial data could overestimate or underestimate the actual efficacy of the drug.

Another key uncertainty lies in trial costs. The cost of clinical trials is not fixed and can fluctuate due to factors such as recruitment delays, additional testing requirements, or manufacturing challenges. For this reason, I set Phase I costs to vary between \$10 million and \$50 million, instead of assuming a strict \$30 million.

To evaluate the impact of uncertainty, I conducted 1,000 to 5,000 Monte Carlo simulations, generating a distribution of possible financial outcomes instead of relying on a single expected value. The distribution of optimal decisions using 5000 simulations shown in the histogram below.

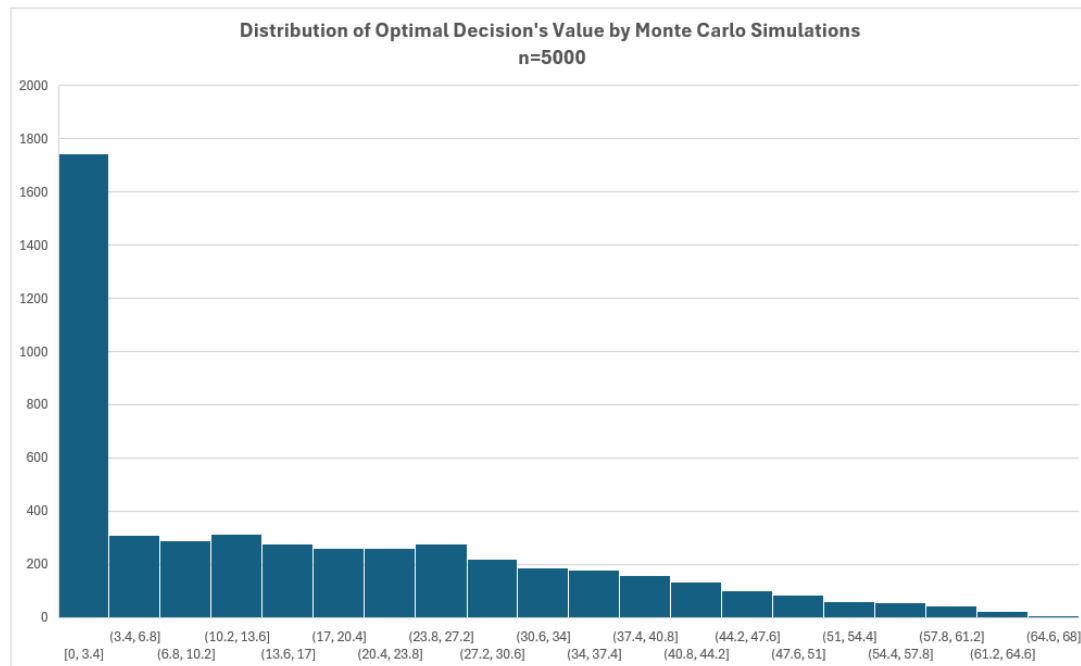


Figure 2

The results show that while some scenarios lead to an expected value of \$0, indicating that not licensing would have been the better choice, the majority of cases still yield a positive expected value, favoring the decision to license Davanrik as shown in summary statistics in figure 3.

Monte Carlo Simulations (n = 5000)	
Average Value of Optimal Decision	16.06
Frequency of “License”	3573
Frequency of “Don’t License”	1427

Figure 3

The Monte Carlo simulation strengthens the credibility of the initial decision tree analysis by demonstrating that Merck is likely to generate a positive return under a wide range of conditions. This confirms that while uncertainty exists, the potential profit still outweighs the risk.

Results

The results from both the deterministic decision tree analysis and the Monte Carlo simulations confirm that licensing Davanrik is the better financial decision for Merck. The decision tree analysis that assumes fixed probabilities and costs shows that licensing Davanrik has an expected value of \$13.98 million, while

rejecting the offer results in an expected value of \$0. Under these conditions, the potential revenue from a successful launch outweighs the costs and risks of clinical trials, making licensing the optimal choice. However, this model does not account for real-world uncertainties, such as variations in success rates or unexpected trial costs.

To incorporate uncertainty, I ran 1,000 to 5,000 Monte Carlo simulations, allowing trial success rates and costs to vary within realistic industry ranges. The results show that while some simulations lead to an expected value of \$0, meaning not licensing would have been preferable in those cases, the majority of scenarios still result in a positive expected value, supporting the decision to license Davanrik.

At 5,000 simulations, the expected value stabilizes around \$16 million, higher than the deterministic estimate, reinforcing that licensing remains financially favorable even under uncertainty. The simulation results also show that 3,573 cases favor licensing, while only 1,427 cases suggest rejecting the drug, confirming that licensing is the better decision in most cases.

Recommendation

Based on the results, I recommend that Merck proceed with licensing Davanrik. Since I assume Merck is a risk-neutral decision maker, the optimal decision should be based solely on expected value. The deterministic decision tree analysis shows that licensing Davanrik yields a positive expected value of \$13.98 million. Additionally, the average expected value under Monte Carlo simulations remains around \$16 million across 5,000 simulations, reinforcing that licensing remains the better decision overall.

However, financial projections alone do not guarantee success. Merck must actively manage risks and optimize its approach to ensure Davanrik's profitability. To maximize the chances of success, Merck should focus on three key areas: clinical trial management, cost control, and market readiness. First, closely monitoring clinical trial progress is crucial. Early identification of trial inefficiencies, patient response variations, and potential safety concerns will allow Merck to adjust strategies and improve the likelihood of regulatory approval. Since unexpected trial failures remain a financial risk, the company must ensure that trial design, patient recruitment, and regulatory compliance remain strong throughout all phases.

Second, controlling development costs will be essential to maintaining profitability. The Monte Carlo simulations highlight the impact of fluctuating costs, particularly in Phase I and Phase II trials. Merck should negotiate cost-effective trial management, optimize resource allocation, and prepare for potential overruns. This approach will ensure that even if costs rise slightly above initial estimates, the financial return remains favorable.

Finally, Merck must develop a strong market entry strategy to maximize revenue potential. Pricing, reimbursement policies, and competition will influence the drug's commercial success. A competitive pricing strategy should balance profitability with affordability to ensure broad market adoption. Additionally, securing regulatory approvals efficiently and establishing strong partnerships with healthcare providers and insurers will further enhance Davanrik's market potential.

Conclusion

This analysis determines that, on average, licensing Davanrik remains the optimal decision for Merck. The deterministic decision tree model estimates an expected value of \$13.98 million, while the Monte Carlo simulations stabilize around \$16 million on average after 5,000 iterations. Although some scenarios in the Monte Carlo simulations result in an expected value of \$0, the majority of outcomes still produce a positive expected value, reinforcing the decision to license Davanrik.

Assuming Merck is a risk-neutral decision-maker, Merck will base its decision solely on expected value. Given the financial projections, licensing remains the most favorable option as long as conditions remain stable. Unless there is a significant increase in trial costs, lower-than-expected success rates, or major regulatory challenges, the optimal decision continues to be to license Davanrik. While uncertainty exists, the results show that on average, the financial benefits of investing in drug development or licensing Davanrik outweigh the risks, making licensing the optimal decision for Merck & Co.

APPENDIX A

Additional Tables

Introduction

This appendix contains additional tables referenced in the main report. These visuals provide further insights and support the data and analysis presented. Each table is labeled and referenced appropriately within the report for easy cross-referencing.

Cost and Probability of Each Phase

Phase I		
Cost (in millions)		30
Chance of Success	$P(S)$	0.6
Change of Failure	$P(F)$	0.4

Phase II		
Cost (in millions)		40
Chance of efficacious for depression	$P(D S)$	0.1
Chance of efficacious for weight loss	$P(WL S)$	0.15
Chance of efficacious for both depression and weight loss	$P(B S)$	0.05
Change of Failure	$P(F S)$	0.7

Phase III		
Cost given efficacious for depression only (in millions)	200	
Chance of Successful outcome for depression only	$P(D D)$	0.85
Chance of Failure	$P(F D)$	0.15
Cost given efficacious for weight loss only (in millions)	150	
Chance of Successful outcome for weight loss only	$P(WL WL)$	0.75
Change of Failure	$P(F WL)$	0.25
Cost of indication test given efficacious for both depression and weight loss (in millions)	500	
Chance of successful outcome for depression only	$P(D B)$	0.15
Chance of successful outcome weight loss only	$P(WL B)$	0.05
Chance of successful outcome both depression and weight loss	$P(B B)$	0.07
Chance of complete failure	$P(F B)$	0.1

Revenue & Profit Forecast from Davanrik

	Revenue from Launch (in millions)	Cost to Launch (in millions)	Profit (in millions)
Launch Depression Only	1200	250	950
Launch Weight-Loss Only	345	100	245
Launch Both	2250	400	1850

APPENDIX B

Detailed Explanation of Excel Workbook: Zahrani_Hanum-2. xlsx

Introduction

The following appendix provides a detailed explanation of the Excel workbook used for the analysis done in this business report. The appendix will walk you through the formulas and functions used and the process undertaken. Understanding these details will give the reader a clearer picture of the underlying calculations and methodologies that support my findings.

Part A: Decision Tree without Monte Carlo Simulations

Before constructing the decision tree, the first step is to compile all relevant information in a structured table:

Given Information about Cost and Probability

Information Given			
Phase	Detail	Value (cost in millions)	Symbol
Phase I	Cost	30	
	Chance of Success	0.6	P(S)
	Chance of Failure	0.4	P(F)
Phase II	Cost	40	
	Chance of efficacious for depression	0.1	P(D S)
	Chance of efficacious for weight loss	0.15	P(WL S)
	Chance of efficacious for both depression and weight loss	0.05	P(B S)
	Change of Failure	0.7	P(F S)
Phase III	Cost given efficacious for depression only	200	
	Chance of Successful outcome for depression only	0.85	P(D D)
	Chance of Failure	0.15	P(F D)
	Cost given efficacious for weight loss only	150	
	Chance of Successful outcome for weight loss only	0.75	P(WL WL)
	Chance of Failure	0.25	P(F WL)
	Cost of indication test given efficacious for both depression and weight loss	500	
	Chance of successful outcome for depression only	0.15	P(D B)
	Chance of successful outcome weight loss only	0.05	P(WL B)
	Chance of successful outcome both depression and weight loss	0.7	P(B B)
	Chance of complete failure	0.1	P(F B)

Given Information about Revenue & Cost of Launch

	Revenue from Launch	Cost to Launch	Profit from Launch
Launch Depression only	1200	250	950
Launch Weight-loss Only	345	100	245
Launch Both	2250	400	1850

Once the given information is clearly laid out, the next step is to calculate the expected value of licensing Davanrik using backward induction, starting from Phase III:

Step 1: Calculating the Expected Value for Phase III

STEP 1 (Phase III)	
P (D D)	0.85
P (F D)	0.15
\$S =	750
\$F =	-200
\$Do Nothing	-70
EV Invest in phase III	607.5
Value (Optimal Decision) =	607.5
Optimal Decision =	Invest
P(WL WL)	0.75
P(F WL)	0.25
\$S =	95
\$F =	-150
\$Do Nothing	-70
EV Invest in Phase III=	33.75
Value (Optimal Decision) =	33.75
Optimal Decision =	Invest
P(D B)	0.15
P(WL B)	0.05
P(B B)	0.7
P(F B)	0.1
\$D	450
\$WL	-255
\$B	1350
\$F =	-500
\$Do Nothing	-70
EV Invest in Phase III	949.75
Value (Optimal Decision) =	949.75
Optimal Decision =	Invest

The first step in constructing the decision tree was to determine the expected value of investing in Phase III, given different possible outcomes from Phase II. The profitability of investing in Phase III depends on whether the drug was found effective for depression only, weight loss only, or both conditions in Phase II. If Phase III trials fail, Merck incurs a loss equivalent to the costs of the trials.

In the Excel workbook, the expected value (EV) for each Phase III scenario was computed using the following general formula:

$$EV = P(S) \times S + P(F) \times F$$

Where:

- EV = Expected Value of investing in Phase III
- P(S) = Probability of Phase III success
- P(F) = Probability of Phase III failure
- S = Profit from launching the drug if successful
- F = Financial loss if the trial fails

Potential profit (\$S) for each scenario is calculated as total revenue from launch minus development costs, we define:

$$S = \text{Profit from Launch} - \text{Phase III Costs}$$

For Failure:

$F = -\text{Phase III Costs}$

If Merck chooses to do nothing, it avoids further costs but incurs a loss equal to the sum of prior sunk costs:

$EV(\text{Do Nothing}) = -(\text{cost of Phase I} + \text{Cost of Phase II})$

$EV(\text{Do Nothing}) = -(30+40) = -70$

I determine the optimal decision by comparing the expected value of investing in phase III with Do Nothing. If $EV(\text{Invest}) > EV(\text{Do Nothing})$, then the optimal choice is to proceed with Phase III.

Case 1: Expected Value for Phase III Given Phase II Shows Effectiveness for Depression Only

If Phase II shows effectiveness for depression only, the Phase III outcomes are:

- Success (85%) → The drug is approved, generating \$750 million in profit.
- Failure (15%) → The trial fails, and Merck loses \$200 million in costs.

The expected value is determined using the formula:

$$EV(D|D) = P(D|D) \times S + P(F|D) \times F$$

Where:

- S (Success Value) = Gross Profit from Launch Depression Only - Cost of Phase III for Depression Only $S=950-200=750$
- F (Failure Value) = - Cost of Phase III for Depression Only $F=-200$

Thus, the expected value calculation in Excel is:

$$EV(D|D) = (0.85 * 750) + (0.15 * -200) = 607.5$$

To determine the optimal decision, I compare the EV with the "Do Nothing" option:

$\text{Do Nothing} = -(\text{Cost of Phase I} + \text{Cost of Phase II}) = -(30 + 40) = -70 \text{ million}$

I use the MAX() function in Excel to select the optimal decision:

=MAX(Cell (Do Nothing), Cell(Expected Value of Phase III))

For each scenario, I used the IF() function to decide whether to proceed with investment. Since $EV(D|D) = 607.5M$ and Do Nothing = -70M, the formula applied in Excel is:

=IF(607.5 > -70, "Invest", "Do Nothing")

Resulting in “Invest” as an optimal choice, meaning that given Phase II depression only, Merck should continue to Invest in Phase III since Phase III will generate positive expected value rather than do nothing.

Case 2: Expected Value for Phase III Given Phase II Shows Effectiveness for Weight Loss Only

If Phase II demonstrates effectiveness for weight loss only, the outcomes in Phase III are:

- Success (75%) → The drug is approved, generating \$95 million in profit.
- Failure (25%) → The trial fails, and Merck loses \$150 million in costs.

The expected value for this scenario is calculated using the formula:

$$EV(WL | WL) = P(WL | WL) \times S + P(F | WL) \times F$$

Where:

- S (Success Value) = Gross Profit from Launch (Weight Loss Only) - Cost of Phase III for Weight Loss Only = $245 - 150 = 95$
- F (Failure Value) = - Cost of Phase III for Weight Loss Only = -150

Thus, the expected value calculation in Excel is:

$$EV(WL | WL) = (0.75 \times 95) + (0.25 \times -150) = 33.75$$

To determine the optimal decision, I compare the EV of investing in Phase III with the Do Nothing option:

$$EV(\text{Do Nothing}) = -(30 + 40) = -70 \text{ million}$$

In Excel, I use the MAX() function to select the optimal decision:

=MAX(Cell (Do Nothing), Cell(Expected Value of Phase III))

For each scenario, I applied the IF() function to determine whether to proceed with investment:

=IF(33.75>-70,"Invest","DoNothing")

Since $33.75 > -70$, the function returns: "Invest".

Thus, given Phase II shows effectiveness for weight loss only, Merck should continue to invest in Phase III, as the expected value remains positive and higher than the Do Nothing option.

Case 3: Expected Value for Phase III Given Phase II Shows Effectiveness for Both Conditions

If Phase II demonstrates effectiveness for both depression and weight loss, Phase III will require specialized dual indication testing, leading to four possible outcomes:

1. Success for Depression Only (15%) → The drug is approved for depression only, generating \$450 million in profit.
2. Success for Weight Loss Only (5%) → The drug is approved for weight loss only, generating -255 million (a loss due to costs exceeding revenue).
3. Success for Both Indications (70%) → The drug is approved for both conditions, generating \$1,350 million in profit.
4. Complete Failure (10%) → The trial fails, and Merck loses \$500 million in costs.

The expected value for this scenario is calculated using the formula:

$$EV(B|B) = P(D|B) \times SD + P(WL|B) \times SWL + P(B|B) \times SB + P(F|B) \times F$$

Where:

- S_D (Depression Only Success Value) = Gross Profit from Launch (Depression Only) - Cost of Dual Indication Testing

$$SD = 950 - 500 = 450$$

- S_WL (Weight Loss Only Success Value) = Gross Profit from Launch (Weight Loss Only) - Cost of Dual Indication Testing

$$SWL = 245 - 500 = -255$$

- S_B (Both Conditions Success Value) = Gross Profit from Launch (Both) - Cost of Dual Indication Testing

$$SB=1850-500=1350$$

- F (Failure Value) = - Cost of Dual Indication Testing

$$F=-500$$

Thus, the expected value calculation in Excel is:

$$EV(B|B)=(0.15 \times 450)+(0.05 \times -255)+(0.7 \times 1350)+(0.1 \times -500)=949.75$$

To determine the optimal decision, I compare the EV of investing in Phase III with the Do Nothing option:

$$EV(\text{Do Nothing})=-(30+40)=-70 \text{ million}$$

In Excel, I use the MAX() function to select the optimal decision:

$$=MAX(\text{Cell (Do Nothing)}, \text{Cell(Expected Value of Phase III)})$$

For each scenario, I applied the IF() function to determine whether to proceed with investment:

$$=IF(\text{Cell (Do Nothing)} > \text{Cell(Expected Value of Phase III)}, "Invest", "DoNothing")$$

Since $949.75 > -70$, the function returns: "Invest".

Thus, given Phase II shows effectiveness for both depression and weight loss, Merck should continue to invest in Phase III, as the expected value remains positive and higher than the Do Nothing option.

Step 2: Calculating the Expected Value for Phase II

STEP 2 (Phase II)	
P (D S)=	0.1
P(WL S)=	0.15
P(B S)=	0.05
P(F S)=	0.7
\$D=	567.5
\$WL=	-6.25
\$B=	909.75
\$F=	-40
\$Do Nothing=	-30
EV Invest in Phase II=	73.3
Value (Optimal Decision)=	73.3
Optimal Decision=	Invest

The next step in constructing the decision tree is to determine the expected value of investing in Phase II, given the different possible outcomes. Phase II trials assess Davanrik's efficacy in treating depression, weight loss, or both conditions. The results from Phase II determine whether the drug advances to Phase III or is discontinued due to failure. If Merck proceeds with Phase II and the trials fail, the company incurs a loss equivalent to the costs of the Phase II trials.

To calculate the expected value (EV) for investing in Phase II, we use the formula:

$$EV = P(D | S) \times SD + P(WL | S) \times SWL + P(B | S) \times SB + P(F | S) \times F$$

Where:

- EV = Expected Value of investing in Phase II
- P(D|S), P(WL|S), P(B|S), P(F|S) = Probabilities of Phase II outcomes (Depression Only, Weight Loss Only, Both, or Failure)
- S_D, S_WL, S_B = Expected value of proceeding to Phase III from each outcome
- F = Financial loss if the trial fails

The profit values (\$S) for each outcome come from the Phase III expected values, previously computed:

- S_D (Depression Only Success Value) = 607.5 (given information)
- S_WL (Weight Loss Only Success Value) = 33.75 (given information)
- S_B (Both Conditions Success Value) = 949.75 (given information)
- F (Failure Value) = -40 (cost of Phase II without return)

Using the probabilities and corresponding values:

$$EV(\text{Invest in Phase II}) = (0.1 \times 607.5) + (0.15 \times 33.75) + (0.05 \times 949.75) + (0.7 \times -40)$$

$$EV(\text{Invest in Phase II}) = 60.75 + 5.06 + 47.49 - 28$$

$$EV(\text{Invest in Phase II}) = 73.3$$

To determine whether Merck should invest in Phase II, I compare the expected value of investing with the Do Nothing option:

$$EV(\text{Do Nothing}) = -(\text{Cost of Phase I}) = -30$$

In Excel, I use the MAX() function to select the optimal decision:

$$=MAX(\text{Cell (Do Nothing)}, \text{Cell (Expected Value of Phase II)})$$

For each scenario, I applied the IF() function to determine whether to proceed with investment:

$$=IF(\text{Cell (Do Nothing)} > \text{Cell (Expected Value of Phase II)}, \text{"Invest"}, \text{"Do Nothing"})$$

Since $73.3 > -30$, the function returns: "Invest".

Thus, given the probabilities and costs associated with Phase II, Merck should continue to invest in Phase II, as the expected value is positive and higher than the Do Nothing option.

Step 3: Calculating the Expected Value for Phase I (Licensing Decision)

STEP 3 (Phase I)	
P (S)=	0.6
P (F)=	0.4
\$S=	43.3
\$F=	-30
Do Nothing=	0
EV Invest in Phase I=	13.98
Value (Optimal Decision)=	13.98
Optimal Decision=	Invest

The final step in the decision tree analysis is determining the expected value of investing in Phase I, which ultimately represents Merck's choice to license Davanrik or reject the offer. Since Phase I is the first stage of clinical trials, its primary purpose is to test the drug's safety before proceeding to later phases. If the trial is successful, the drug moves on to Phase II with a chance of proving its effectiveness. If Phase I fails, Merck loses the initial investment cost and cannot proceed further.

To calculate the expected value (EV) of investing in Phase I, I use the standard formula:

$$EV = P(S) \times S + P(F) \times F$$

Where:

- EV = Expected Value of investing in Phase I
- P(S), P(F) = Probabilities of Phase I success or failure
- S = Expected value from proceeding to Phase II (previously calculated in Step 2)
- F = Financial loss if the trial fails

From the previous step, we use:

- S (Success Value) = 73.3 (Expected value from investing in Phase II)
- F (Failure Value) = -30 (Cost of Phase I if the trial fails)

Using the given probabilities:

$$EV(\text{Invest in Phase I}) = (0.6 \times 73.3) + (0.4 \times -30)$$

$$EV(\text{Invest in Phase I}) = 43.3 - 12$$

$$EV(\text{Invest in Phase I}) = 13.98$$

To decide whether Merck should license Davanrik, I compare the expected value of investing with the Do Nothing option:

$$EV(\text{Do Nothing}) = 0$$

Since Merck has not yet committed any financial resources, choosing not to license results in zero expected value, meaning no gain but also no loss.

In Excel, I apply the MAX() function to determine the optimal decision:

$$=MAX(\text{Cell (Do Nothing)}, \text{Cell (Expected Value of Phase I)})$$

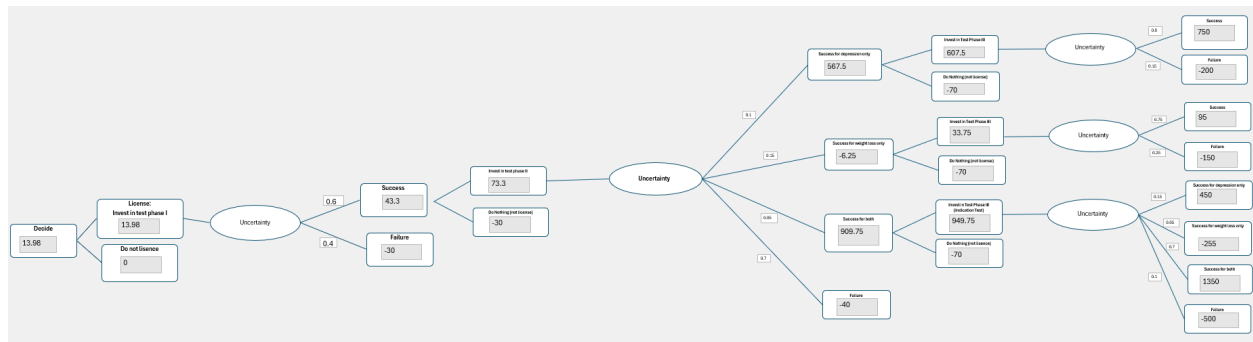
For the final licensing decision, I use the IF() function to confirm whether to proceed:

$$=IF(\text{Cell (Expected Value of Phase I)} > \text{Cell (Do Nothing)}, "Invest", "DoNothing")$$

Since $13.98 > 0$, the function returns: "Invest".

Thus, Merck should proceed with licensing Davanrik, as the expected value is positive and greater than rejecting the offer. This confirms that under the given probabilities and cost assumptions, licensing Davanrik is the financially optimal decision.

Visualizing the Decision Tree in Excel



After calculating the expected values for each phase, I created a decision tree visualization in Excel to clearly represent the possible choices, probabilities, and expected outcomes. The decision tree helps illustrate the step-by-step process of moving through Phase I, Phase II, and Phase III, allowing for a structured comparison of the financial impact of each decision.

To construct the decision tree, I used Excel shapes and lines to create the branching structure. I inserted text boxes to label each node with its corresponding expected value and probability. Instead of manually inputting values, I linked each text box to the respective calculated cell in the Excel table using the = (equals sign) function. This approach ensures that if any values in the table are updated, the decision tree automatically updates to reflect those changes.

For each decision point, I visually represented the choice between investing or doing nothing and connected these to the corresponding probabilities of success and failure. The final expected values at each stage determine the optimal decision, which is highlighted in the decision tree. The visualization not only simplifies the interpretation of complex calculations but also enhances the ability to analyze different scenarios dynamically by adjusting input values in the Excel workbook.

Part B: Monte Carlo Simulations

To enhance the robustness of the decision-making process, I incorporated Monte Carlo simulations to introduce uncertainty into key parameters. I began by copying all calculations and formulas from Part A into a new Excel workbook, naming it "Part B" to serve as the foundation for the simulation process. In real-world scenarios, success probabilities and costs fluctuate due to unpredictable factors such as

regulatory hurdles, patient variability, and trial conditions. Therefore, instead of assuming fixed values, in Part B, I allowed these parameters to vary within a reasonable range to simulate different possible outcomes.

The key parameters modified for uncertainty include:

1. Probability of success in Phase I: Adjusted within a range of 40% to 80%.
2. Cost of failure in Phase I (\$F): Allowed to fluctuate between \$10 million and \$50 million.
3. Probability of Phase II showing efficacy for both depression and weight loss: Set within 0% to 10% to reflect uncertainty in clinical trial results.

To implement randomness within these constraints, I used the RAND() function in Excel:

$$=\text{Min}+\text{RAND}() \times (\text{Max}-\text{Min})$$

This formula ensures that each simulation selects a random value for the specified parameter within its defined range.

Since all expected values (EV), Value of Optimal Decision, and Optimal Decision (Invest or Not) were already structured in the Excel model, these values automatically updated whenever the probabilities and costs changed. Additionally, since the decision tree was linked to these values, all figures within the decision tree also adjusted dynamically when uncertainty was introduced.

Next, I set up Monte Carlo simulations to analyze how the decision to license Davanrik varied under different simulated conditions.

I first created a table that stored the key variables for each simulation:

Value (Optimal Decision)=	14.71261563
Optimal Decision =	Invest
Rand() =	0.065742962

Where:

1. Value of Optimal Decision: Corresponds to the expected value of licensing Davanrik in Phase I.

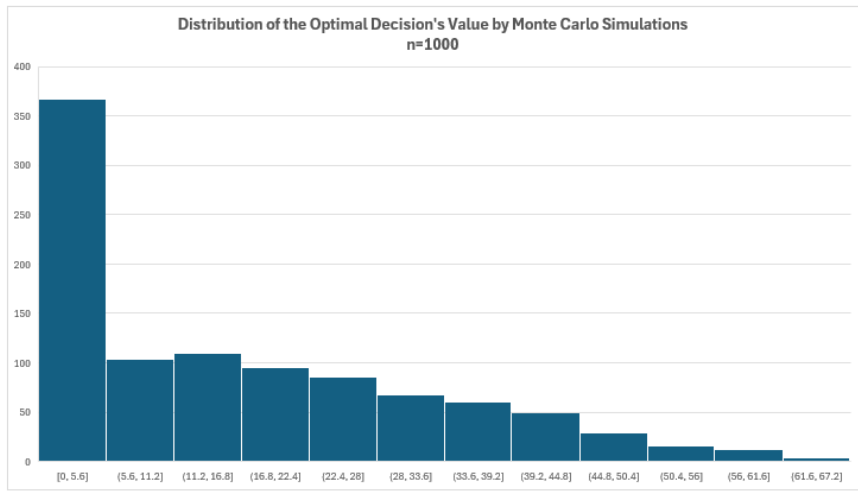
2. Investment Decision: Determined using an IF() function, applied as follows:
=IF(Cell (Value of Optimal Decision)>0,"Invest","Nothing")=IF(\text{Cell (Value of Optimal Decision)} > 0, "Invest", "Nothing")=IF(Cell (Value of Optimal Decision)>0,"Invest","Nothing")
3. Random Number: Generated using RAND(), which is used to trigger recalculations in the Monte Carlo process.

Once the setup was complete, I conducted Monte Carlo simulations using What-If Analysis in Excel. The process followed these steps:

1. Creating the Simulation Table
 - The first column contained simulation numbers (e.g., 1 to 1000).
 - The second column stored the Value of Optimal Decision, dynamically linked to Phase I's EV cell.
 - The third column displayed the Investment Decision using the IF() function: =IF(Value of Optimal Decision>0,"Invest","Nothing")=IF(\text{Value of Optimal Decision} > 0, "Invest", "Nothing")=IF(Value of Optimal Decision>0,"Invest","Nothing")
2. Applying What-If Analysis
 - I selected the Monte Carlo simulation table and navigated to What-If Analysis → Data Table.
 - The Row Input Cell was left empty.
 - The Column Input Cell was set to the random probability variable in the parameter table.
 - Clicking "OK" ran the simulation across all instances.
3. Increasing the Simulation Sample Size
 - After completing 1,000 simulations, I repeated the same process for 2,000, 3,000, 4,000, and 5,000 simulations as shown in figure below.
 - This allowed me to observe how the decision trends stabilized as the number of trials increased.

Monte Carlo Simulations											
n=1000	17.44340899	Invest	n=2000	17.443409	Invest	n=3000	17.443409	Invest	n=4000	17.443409	Invest
1	36.0660968	Invest	1	46.027007	Invest	1	3.034932778	Invest	1	0	Nothing
2	27.96404372	Invest	2	152970	Invest	2	31.071795	Invest	2	50.97338174	Invest
3	11.46510777	Invest	3	43.537619	Invest	3	10.01027982	Invest	3	5.957051634	Invest
4	0.475313432	Invest	4	24.412095	Invest	4	22.274281	Invest	4	60.97985216	Invest
5	5.944989314	Invest	5	0.000000	Nothing	5	0	Nothing	5	32.30013389	Invest
6	0	Nothing	6	33.230522	Invest	6	18.09131671	Invest	6	18.77686479	Invest
7	9.87539949	Invest	7	0.272816	Invest	7	48.11070666	Invest	7	50.40346953	Invest
8	24.45828054	Invest	8	47.422722	Invest	8	37.20877742	Invest	8	22.33323904	Invest
9	0	Nothing	9	23.296434	Invest	9	53.01486781	Invest	9	0	Nothing
10	11.56242543	Invest	10	16.508593	Invest	10	0	Nothing	10	18.59435441	Invest
11	2.624323012	Invest	11	20.853615	Invest	11	13.0438623	Invest	11	19.66251815	Invest
12	0	Nothing	12	14.363033	Invest	12	12.819555	Invest	12	23.88200062	Invest
13	0	Nothing	13	0.000000	Nothing	13	0	Nothing	13	26.03224674	Invest
14	21.39450389	Invest	14	11.996085	Invest	14	0	Nothing	14	23.32762956	Invest
15	2.638999279	Invest	15	57.312571	Invest	15	17.75648324	Invest	15	0	Nothing
16	2.304917724	Invest	16	4.802933	Invest	16	48.45756486	Invest	16	25.57407499	Invest
17	22.24795148	Invest	17	16.195585	Invest	17	50.13426395	Invest	17	16.18257914	Invest
18	22.85949342	Invest	18	15.945114	Invest	18	3.572442402	Invest	18	0	Nothing
19	26.4771344	Invest	19	0.000000	Nothing	19	0	Nothing	19	0	Nothing
20	14.45106844	Invest	20	13.437720	Invest	20	48.75510272	Invest	20	0	Nothing
21	13.57880331	Invest	21	0.000000	Nothing	21	26.14773901	Invest	21	20.21982597	Invest
22	7.789593418	Invest	22	42.389994	Invest	22	0	Nothing	22	46.46916595	Invest
23	17.67480159	Invest	23	34.328464	Invest	23	10.93780548	Invest	23	24.55956297	Invest
24	12.16321723	Invest	24	31.363359	Invest	24	0	Nothing	24	3.002521939	Invest
25	12.36470586	Invest	25	0.000000	Nothing	25	33.24874093	Invest	25	0.440056207	Invest
26	40.80562582	Invest	26	9.135729	Invest	26	0	Nothing	26	0	Nothing
27	21.78831893	Invest	27	57.703948	Invest	27	12.86779241	Invest	27	0	Nothing
28	35.28034608	Invest	28	4.004263	Invest	28	5.28086845	Invest	28	0	Nothing
29	6.770940107	Invest	29	0.574746	Invest	29	0	Nothing	29	10.37423115	Invest
30	0	Nothing	30	9.286422	Invest	30	43.27371968	Invest	30	0	Nothing
31	0	Nothing	31	0.000000	Nothing	31	21.06360458	Invest	31	0	Nothing
32	14.81959895	Invest	32	0.365766	Invest	32	17.30069337	Invest	32	7.033983066	Invest
33	44.37437034	Invest	33	21.352948	Invest	33	14.57660511	Invest	33	28.30661562	Invest
34	29.869354	Invest	34	23.601653	Invest	34	12.4673915	Invest	34	0	Nothing
35	0.201821658	Invest	35	12.178952	Invest	35	9.858667195	Invest	35	3.688044576	Invest
36	0	Nothing	36	0.000000	Nothing	36	21.67213882	Invest	36	6.139486015	Invest
37	0.570696792	Invest	37	7.264240	Invest	37	31.4063716	Invest	37	0	Nothing

After running the Monte Carlo simulations, I created histograms to visualize how the optimal decision’s value was distributed across different simulations.



Steps to create histograms in Excel:

1. Select the Value of Optimal Decision column from the simulation results.
2. Navigate to Insert → Recommended Charts → Histogram.
3. Generate a frequency distribution histogram for each batch (1,000, 2,000, 3,000, 4,000, and 5,000 simulations).

These visualizations helped assess the distribution of possible financial outcomes and how often investing was the preferred decision.

Then, to gain deeper insights into the simulation outcomes, I calculated key statistical measures for each simulation batch. Below are the Excel formulas used to compute these measures for 1,000 simulations (similar formulas were used for larger sample sizes). The formulas reference the range C15:C11004, which represents all the simulated values of the optimal decision across 1,000 trials :

Average Value of optimal decision -> =AVERAGE(C15:C11004)

Frequency of “invest” -> =COUNTIF(C15:C11004,">0")

Frequency of Nothing -> =COUNTIF(C15:C11004,"=0")

Mean -> =AVERAGE(C15:C11004)

Standard error -> =STDEV.S(C15:C11004)/(SQRT(1000))

Median -> =MEDIAN(C15:C11004)

Mode -> =MODE.SNGL(CN5:CN2004)

Standard Deviation -> =STDEV.S(C15:C11004)

Sample Variance -> =VAR.S(C15:C11004)

Kurtosis -> =KURT(C15:C11004)

Skewness -> =SKEW(C15:C11004)

Range -> =MAX(C15:C11004)-MIN(C15:C11004)

Minimum -> =MIN(C15:C11004)

Maximum -> =MAX(C15:C11004)

Sum -> =SUM(C15:C11004)

Count -> =COUNT(C15:C11004)